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Responding to upstream uncertainty: How does supplier risk-taking affect client ESG performance?

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ABSTRACT

With environmental, social, and corporate governance (ESG) issues gaining global prominence, it is increasingly recognized that sustainable practices must extend beyond individual organizations to encompass their entire supply chains. This study investigates the impact of supplier risk-taking on the ESG performance of client firms, highlighting how supply chain transmission influences corporate sustainability initiatives. Using data from Chinese listed firms from 2009 to 2023, our results reveal that supplier risk-taking positively influences the ESG performance of client firms. This positive effect is found to be progressively strengthened as corporate reputation and supplier dependence increase but gradually weakened with rising investor confidence. These findings not only deepen the understanding of the supply chain's role in shaping corporate ESG performance but also highlight ESG as a proactive strategy for client firms to anticipate and mitigate risks transmitted from suppliers. This offers valuable insights for managers and policymakers seeking to enhance sustainable practices through effective supply chain risk management.

1. Introduction

With the growing prominence of environmental, social, and corporate governance (ESG) issues globally, corporate stakeholders are placing greater emphasis on its ESG performance, which is directly linked to sustainable development and long-term value. For example, investors integrate corporate ESG performance into their investment evaluation criteria (Chen and Xie, 2022), while consumers who prefer green products actively seek environmentally friendly goods to minimize environmental pollution (Yang et al., 2019). In response to stakeholder expectations, many firms have integrated ESG into their strategic objectives to tackle environmental and social challenges (Ameer and Othman, 2012; Schrettle et al., 2014). Meanwhile, scholars are increasingly exploring how firms balance profit goals with social responsibility in complex market environments, establishing ESG as a prominent research focus (Liu et al., 2023).

Existing literature has extensively examined the determinants of corporate ESG performance from internal and external perspectives, such as shareholder structure (Barko et al., 2022; Wang et al., 2023a,b), corporate governance (Lagasio and Cucari, 2019; Wu et al., 2023), board structure (Shu et al., 2024), carbon emission policies (Shu and Tan, 2023), green credit policies (Cui et al., 2024), and monetary policies (Abakah et al., 2024). However, in the era of globalization and digitalization, the supplier-client relations become increasingly complex and interconnected, risks arising from any node in the supply chain can have far-reaching consequences for the entire network (Kim et al., 2015). As such, the stability, transparency, and sustainability of supply chains play a crucial role in shaping corporate social responsibility performance (Liu et al., 2023; Lin, 2023).

Recent research shows that downstream digitalization can enhance suppliers' green productivity and performance, providing valuable insight into how firm behaviors transmit along the supply chain (Jin et al., 2025; Zhang et al., 2024). More relatedly, a few studies have examined the impact of supplier misconduct on client ESG performance (Li and Cuervo-Cazurra, 2024; Hu et al., 2024), in which "misconduct" represents a deviation from legal or ethical norms—such as accounting fraud or environmental violations—whose expected consequences are predominantly negative (Greve et al., 2010; Karpoff et al., 2005). These unethical behaviors are typically revealed ex post through regulatory enforcement and event records (e.g., accounting and auditing enforcement releases; environmental penalties), as well as publicly available sources such as media reports or litigation (Karpoff et al., 2008; Dyck

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et al., 2010). In this regard, when upstream suppliers are identified with misconduct behaviors, client firms tend to passively take reactive and remedial responses (e.g., enhanced disclosure, event-driven CSR initiatives) aimed at short-term reputational repair and regulatory compliance (Hu et al., 2024; Karpoff et al., 2008; Dyck et al., 2010; Li et al., 2023).

By contrast, "strategic risk-taking" refers to value-seeking strategic choices made under uncertainty, aimed at achieving long-term objectives like market share growth, technological advancement, or profitability (John et al., 2008; Hoskisson et al., 2017), often resulting in an increase in the variance of operational or financial outcomes. Unlike corporate misconduct, which is normally found ex post, risk-taking can be anticipated ex ante through policy/structural indicators observable to stakeholders—such as the volatility of return on assets, leverage and debt maturity structure, investment intensity (e.g., R&D), or the extent of foreign expansion (John et al., 2008; Hoskisson et al., 2017). Obviously, such increased uncertainty of upstream suppliers' risk-taking does not necessarily jeopardize client reputational capital and thus lead to the negative financial outcomes of clients.

However, in highly coupled supply chain networks, suppliers' aggressive strategies increase the likelihood of adverse events such as logistics delays or ramp-up variability arising from overseas expansion, higher defect rates and yield losses during intensive R&D transitions, and tightened working capital and delayed deliveries due to excessive asset financialization (Hoskisson et al., 2017; Costello, 2020; Psaromatis et al., 2024). These factors collectively increase lead-time variability, quality inconsistency, and the risk of production stoppages at the client's facilities, thereby creating a clear link between financial risks and factory-level operational disruptions. Beyond potential operational disruptions, these uncertainties can also spill over into reputational risks, which may be transmitted along the supply chain from upstream suppliers to their clients (Hendricks and Singhal, 2005a). For instance, in 2020, to diversify geopolitical risk and serve its North American clients, Taiwan Semiconductor Manufacturing Company (TSMC) announced its plan to build a 5-nm wafer fabrication plant in Arizona, United States. The early "ramp-up" challenges of TSMC—including skilled labor shortages, process delays, and construction slippages—have significantly raised supply chain concerns. Several media outlets noted that with advanced capacity still concentrated in Taiwan, its major clients face greater uncertainty in delivery schedules and capacity allocation.[1,2] As such, to safeguard performance in multi-risk environments and ensure the supply chain resilience, client firms are motivated to proactively take sustained risk management mechanism rather than the event-driven responses to buffer potential disruptions and reassure stakeholders (Lins et al., 2017). The ESG initiatives are considered as one of the most functional governance mechanisms (Ha et al., 2023; Wang et al., 2025), which often include establishing supplier codes of conduct, enhancing audit transparency, and improving safety and environmental standards—measures that help mitigate disruption and reputational spillover risks in advance.

Drawing on stakeholder theory and signaling theory, this study aims to investigate how client firms enhance ESG performance in response to suppliers' risk-taking using data from Chinese listed firms between 2009 and 2023. China offers an ideal set to address our research question. In recent years, ESG performance of Chinese firms has garnered significant attention from numerous research (Wang et al., 2023a,b; He et al., 2023; Long et al., 2024). In addition, China has one of the world's most complex and comprehensive supply chain systems and is among the few countries with a fully integrated industrial chain, which facilitates the transmission of upstream risks to downstream firms. Moreover, the rapid evolution of China's ESG disclosure regulations and the unique governance environment—characterized by strong regulatory oversight and a mix of state and market pressures—further enhance the relevance of the Chinese context for examining the link between supplier risk-taking and client ESG performance.

Our results reveal that an increase in supplier risk-taking positively influences the ESG performance of client firms. We further investigate the underlying mechanisms from multiple stakeholder perspectives and find that the positive relationship between supplier risk-taking on client ESG performance is strengthened by higher client corporate reputation and greater supplier dependence but is weakened by its stronger investor confidence. Moreover, extended analyses show that the positive impact is mainly concentrated in the social (S) and governance (G) dimensions. In addition, specific supplier risk-taking behaviors—such as outward foreign direct investment, R&D investment, and asset financialization—also exert positive effects on client firms' ESG performance. Finally, this relationship is further reinforced by client firms' digitalization and the application of artificial intelligence technologies.

Our study adds to the existing literature in the following ways. First, prior research has proved that corporate engagement in social responsibility activities can be considered as a reactive strategy to respond to supplier misconduct aimed primarily at repairing reputational damage under external stakeholder pressure (Li and Cuervo-Cazurra, 2024; Hu et al., 2024). In the present study, we distinguish corporate risk-taking from misconduct, arguing that risk-taking denotes strategic decisions made by firms under uncertainty, which reflects their willingness to pursue profit opportunities despite potential instability or volatility. This initiative can in fact be assessed in advance through a firm's financial reports and operational conditions (Tian et al., 2022), and thus ESG is served as a proactive risk prevention and absorption strategy for client firms to cope with uncertainties arising from upstream supply chain partners. By broadening the drivers of corporate ESG behavior to encompass upstream supply chain risk factors, this study enriches the understanding of ESG's functional roles and thereby provides a novel perspective for future research in this domain.

Second, research on the consequences of corporate risk-taking has traditionally concentrated on the firm itself—for example, its effects on innovation, performance, and corporate policies (García-Granero et al., 2015; Cain and McKeon, 2016; Díez-Esteban et al., 2017). In this paper, we adopt a stakeholder perspective to integrate corporate risk-taking with supply chain dynamics, examining how upstream suppliers' risk-taking is transmitted through the supply chain to affect downstream client firms' ESG performance, thereby broadening the theoretical boundaries of risk-taking and its external effects.

Third, to investigate how supplier risk influences client firms' ESG performance under varying stakeholder conditions, this study employs a partially linear functional-coefficient model. The results indicate that higher supplier dependence, stronger public reputation, and lower investor confidence amplify the positive impact of supplier risk-taking on client ESG engagement. Notably, under extreme conditions—such as low supplier dependence coupled with high investor confidence—firms may reduce their ESG engagement. This finding uncovers a critical threshold in the ESG transmission mechanism within supply chains, offering more nuanced insights for managers and policymakers seeking to enhance corporate sustainability.

2. Theory and hypothesis development

2.1. Supplier risk-taking and client ESG performance

As one of the most important components of corporate risk, the risk-taking of firms has received a lot of attention from both scholars and practitioners. The risk-taking of firms originates from a company's inclination to engage in adventurous activities such as outward foreign direct investment (OFDI), research and development (R&D) activity,

[1] Source:https://www.reuters.com/technology/tsmcs-us-plant-unlikely-get-latest-chip-tech-before-taiwan-ceo-says-2025-01-17/.

[2] To streamline our focus on supplier's risk-taking behavior instead of misconduct, we have deleted the deforestation case of Unilever company and provided the expanding case of TSMC in the revised manuscript.

and asset financialization. These risk-taking behaviors can lead to breakthrough growth by driving innovation, enabling market expansion, and strengthening competitive advantages (Zhou et al., 2021; Gao et al., 2025), while increasing exposure to downside risks, including earnings volatility, resource misallocation, and strategic failure, thereby heightening operational uncertainty (Vural-Yavaş, 2020).

Although supplier risk-taking is not always directly observable, clients often monitor it indirectly via financial disclosures, delivery performance, and third-party risk ratings—particularly in high-dependence or long-term relationships (Ho et al., 2010; Kalkanci and Plambeck, 2020a). In tightly coupled supply chains, supplier instability can propagate to clients through production stoppages, lead-time variability, and quality degradation, posing tangible operational and reputational threats (Soares et al., 2017; Ivanov, 2020).

According to stakeholder theory, when suppliers' operational stability decreases due to heightened risk-taking, client firms face intensified pressures from multiple stakeholders and thus may strengthen ESG engagement as a preventive measure. First, in modern supply chain systems, client firms maintain close relationships with their suppliers (Sambasivan et al., 2011), and this reliance extends beyond product delivery to shaping stakeholders' perceptions of corporate governance and social responsibility. High supplier risk-taking—such as aggressive foreign expansion, intensive R&D, or excessive asset financialization—can primarily lead to ramp-up variability and delivery instability, increased defect rates and quality volatility, and tightened working capital and supply disruptions, respectively. While these consequences differ in their main channels, they may overlap and jointly heighten uncertainty for client firms, creating uncertainty in supply quality, quantity, and timeliness (Gao et al., 2025; Ivanov and Dolgui, 2020; Brusset and Teller, 2017; Hummels and Schaur, 2013). These disruptions may misalign client performance with stakeholder expectations (Hasan et al., 2022).

To preempt such risks, client firms may strengthen ESG practices as a preventive measure, signaling robust management and a commitment to social welfare. By improving internal controls, transparency, and governance standards (Singh, 2021), they can enhance supply chain risk management and resilience (Wu et al., 2024), thereby mitigating operational instability and potential financial losses (Brusset and Teller, 2017; Germain et al., 2008).

Second, from the perspective of corporate reputation, firms rely on reputational capital to sustain stakeholder trust and shape strategic responses to external uncertainties (Gangi et al., 2020; Neville et al., 2005; Soleimani et al., 2014). When suppliers engage in risk-taking behaviors, client firms face operational uncertainty—such as unstable quality, availability, and delivery—which stakeholders may interpret as evidence of weak supply chain governance (Hartmann and Moeller, 2014; Sreedevi and Saranga, 2017). To avert potential reputational damage, client firms often preemptively strengthen ESG practices, for example by enhancing supplier audits or tightening environmental and social requirements (Lins et al., 2017). Such proactive engagement both mitigates reputational spillovers and signals the firm's commitment to responsible supply chain management and sound governance, even under heightened risk (Bitektine, 2011; Love and Kraatz, 2017).

Third, from the perspective of corporate investors, heightened supplier risk-taking may also adversely affect the market valuation of client firms and, in extreme cases, pose a risk of stock price collapse (Yang and Jiang, 2024; Hertzel et al., 2008; Chen et al., 2021). This can lead client firms to use positive ESG practices as a signal to external investors to demonstrate that they are well-stabilized and worthy of investment. Specifically, before making investment decisions, investors usually need to form a comprehensive picture of a firm's operations, which makes them highly sensitive to vulnerabilities in the supply chain. As such, once information about a supplier's risk-related behavior is perceived by external investors, and due to uncertainty about the future growth and profitability of the client's business, rational investors may reassess a client firm's risk profile based on its dependence on high-risk suppliers, which may postpone their investment and suppress the client firm's market valuation (Barakat et al., 2014). In this context, corporate ESG performance, as an increasingly important factor in investment decision-making, has the potential to attract investor attention and shape perceptions of firm quality (Long et al., 2024). To mitigate the risk of market undervaluation, client firms can use positive ESG signals to demonstrate robust risk control mechanisms (Ding et al., 2024). Such signaling helps reassure investors of the firm's financial stability, mitigate market concerns, and strengthen investment confidence, thereby contributing to an enhanced market value (Nekhili et al., 2021; Cojoianu et al., 2021) and market competitiveness (Zhang et al., 2020; Cheng et al., 2014).

Based on the analysis presented above, we propose Hypothesis 1.

Hypothesis 1. The higher the risk-taking by suppliers, the better the ESG performance of downstream clients.

2.2. The moderating role of supplier dependence

Prior research indicates that the structure of a supply chain significantly influences how supplier risks propagate downstream firms (Bode and Wagner, 2015; Craighead et al., 2007). High supplier dependence amplifies the impact of supplier risk-taking on client firms by limiting their ability to diversify procurement sources and reducing operational flexibility in responding to upstream disruptions (Kamalahmadi and Parast, 2017). This concentrated reliance not only increases the vulnerability of client firms to supplier instability but also heightens stakeholders' sensitivity to risks associated with key suppliers. Stakeholders are more likely to attribute adverse supplier behaviors to deficiencies in the client firm's governance and operational capacity, intensifying reputational and trust-related consequences (Chai and Ngai, 2015; Hendricks and Singhal, 2005b). Consequently, client firms with high supplier dependence exhibit stronger preventive motivations to upstream risk signals.

Moreover, high supplier dependence exacerbates overall supply chain risk and diminishes resilience, as client firms have fewer alternatives to quickly address unexpected supplier failures such as delayed deliveries or substandard products, which can disrupt production and degrade output quality (Leung and Sun, 2021; Chopra and Sodhi, 2014). In this context, the heightened operational risks further motivate client firms to adopt proactive strategies to mitigate potential disruptions and safeguard supply chain stability.

For these reasons, client firms with high supplier dependence tend to both demonstrate proactive ESG performance to signal operational stability, financial soundness, and social responsibility, and intensify risk resilience by adopting enhanced ESG practices (Wang et al., 2024), such as establishing independent risk management committees and ensuring oversight of supplier relations (Zhang et al., 2017). This indicates that supplier dependence strengthens the motivation for client firms to respond to upstream risks through more robust ESG strategies.

In summary, high supplier dependence amplifies the impact of supplier risk-taking and weakens supply chain resilience. Therefore, clients tend to adopt more proactive ESG strategies to mitigate the risks arising from supplier risk-taking. Based on this analysis, we propose Hypothesis 2.

Hypothesis 2. The higher the supplier dependence, the greater the positive effect of supplier risk-taking on client ESG performance.

2.3. The moderating role of client corporate reputation

The reputation of client firms plays a crucial moderating role in the relationship between supplier risk and their ESG performance. On the one hand, firms with superior reputational capital exhibit stronger proactive motivations. As a valuable intangible asset accumulated over time, reputational capital enhances firms' sensitivity to stakeholder perceptions and strengthens their commitment to preserving brand

equity (Bitektine, 2011; Love and Kraatz, 2017). The higher the reputation, the more vigilant firms become toward potential risks that could undermine their credibility. This forward-looking concern renders them particularly sensitive to risk-taking behaviors by upstream suppliers. Accordingly, such firms are more inclined to adopt ESG practices as a preventive governance strategy to contain potential spillovers and reinforce long-term stakeholder trust (Bansal and Shaliastovich, 2010).

On the other hand, high-reputation firms are also subject to heightened external pressure (Zavyalova et al., 2016). Their elevated public visibility makes them more susceptible to media scrutiny and stakeholder criticism when supply chain disruptions occur, intensifying expectations for transparency and responsible oversight (Yu et al., 2017). Consequently, the exposure of supplier risk-taking entails amplified reputational risks for these firms, given their greater stakeholder salience and governance accountability (Hartmann, 2021). Under such scrutiny, reputable firms are further incentivized to adopt ESG as a forward-looking strategy to safeguard their reputation before potential adverse events materialize. In contrast, firms with lower reputational visibility face weaker stakeholder pressure and thus have fewer incentives to engage in proactive ESG adjustments in response to supplier-related risks. Based on the above analysis, we propose hypothesis 3.

Hypothesis 3. The higher the corporate reputation, the greater the positive effect of supplier risk-taking on client ESG performance.

2.4. The moderating role of client investor confidence

As key stakeholders, investors significantly influence a firm's strategic responses (Shahid and Abbas, 2019). According to stakeholder theory, firms adopt ESG practices in part to address investor expectations and maintain external legitimacy (Long et al., 2024). Investor confidence plays a moderating role in this process.

When investor confidence is high, client firms may exhibit a weaker ESG response to supplier risk-taking due to two mechanisms. First, from a stakeholder pressure perspective, optimistic investors tend to exhibit higher risk tolerance (Habib and Hasan, 2017). As a result, they may not exert strong pressure on the firm to take responsive actions but rather maintain moderate expectations regarding its behavior. Second, from a strategic perspective, high investor confidence signals that the firm has already earned stakeholder trust and reputational legitimacy (Firth et al., 2015). When the market has formed a favorable perception of a firm, the marginal benefit of additional signaling efforts declines (Connelly et al., 2011). As a result, firms with abundant reputational capital may be less inclined to invest further in reputation-enhancing activities, such as additional ESG initiatives (Masulis and Reza, 2015). Instead, they may prioritize resource allocation toward financial goals or operational efficiency. Third, from an operational resilience perspective, high investor confidence often reflects strong resource endowments and robust internal capabilities, such as maintaining higher safety stock, diversifying supplier bases, and implementing comprehensive internal control systems (Ivanov and Dolgui, 2020; Tomlin, 2006; Katsaliaki et al., 2022). Such operational buffers enhance firms' ability to absorb and manage potential disruptions from supplier risk-taking, thereby diminishing the need to employ ESG initiatives primarily as symbolic or risk-mitigating signals.

In contrast, when their confidence is low, investors tend to adopt a more risk-averse mindset and become increasingly sensitive to the uncertain consequences of various risk behaviors (Lee et al., 2015), resulting in increased pressure on the client firm to take proactive ESG actions. Consequently, in the context of supplier risk-taking, clients with high investor confidence are less motivated to invest in ESG initiatives to respond to suppliers' risk-taking. As such, we propose Hypothesis 4.

Hypothesis 4. The higher the investor confidence in clients, the weaker the positive effect of supplier risk-taking on client ESG performance.

3. Research design

3.1. Sample and data

Since our research question correlates supply chain data with ESG data, we first take the ESG-disclosing Chinese A-share listed firms from 2009 to 2023 as the base sample and then crossmatch them with the supply chain pairs. Under the disclosure rules of the China Securities Regulatory Commission (CSRC), listed firms are required to report the transaction values and names of their top five suppliers. In practice, supplier names and stock codes are fully disclosed only when the supplier itself is also a listed firm. Consequently, a valid supply chain link can be established only when the supplier appears in the client's disclosed top five list and both the supplier and the client are listed firms. Thus, we retain only firm-year observations where an identifiable supply chain relationship exists in the observation year.

Afterward, following common practice in the literature, we rule out the financial institutions (i.e., banks, insurance companies, securities firms, and trust companies), as well as firms labeled as Special Treatment (ST) or Particular Special Treatment (*ST) from our data (Geng et al., 2024; Wang et al., 2023a,b; Sun et al., 2023). Additionally, to mitigate the potential impact of outliers, observations with substantial missing values or those falling outside the 1st to 99th percentile range were excluded from the regression analysis. As a result, the final non-balanced panel comprises 2800 firm-year observations, which is larger than or consistent with prior studies (Geng et al., 2024; Wang et al., 2023a,b; Sun et al., 2023).

The supply chain data are obtained from the China Stock Market and Accounting Research (CSMAR) database. The risk-taking indicator is constructed based on firms' financial data extracted from the CSMAR database, while ESG index data (Hua Zheng ESG Ratings) are provided by Sino-Securities Index Information Service (Shanghai) Co., Ltd. Additional firm-level characteristics are obtained from the CSMAR, Wind, and Chinese Research Data Services (CNRDS) databases.

3.2. Variables

3.2.1. Dependent variable

ESG performance (*C_ESG*) is considered as the dependent variable in this study, which is evaluated by the Sino-Securities Index ESG Ratings (Hua Zheng ESG index). In the case of China, the Sino-Securities Index ESG Ratings, which covers most of Chinese A-share listed companies, have been extensively applied in prior studies (H. Chen et al., 2025; Li et al., 2023; Ying et al., 2025; Sun et al., 2024). Moreover, recent systematic evidence indicates that domestic ESG ratings—particularly Sino-Securities ESG—offer greater contextual relevance and explanatory power than international ratings when assessing Chinese firms (J. Z. Chen et al., 2025). This is because domestic agencies place greater weight on dimensions aligned with China's institutional environment, including policy orientation, environmental compliance, social stability responsibilities, and governance structures. They also leverage local information sources—such as Chinese-language disclosures, local media reports, and regulatory documents—allowing for a more accurate and interpretable assessment of ESG performance in the Chinese context.

This dataset provides ratings across three dimensions: Environmental (E), Social (S), and Corporate Governance (G), along with an overall ESG rating. Each indicator is classified into nine levels (1–9), with higher levels signifying better ESG performance. We calculate the annual ESG performance by averaging its monthly values over the year.

3.2.2. Independent variable

Following prior research, we use earnings volatility to assess a supplier firm's risk-taking level (John et al., 2008; Li et al., 2013). The independent variable (*S_Risk-taking*) is defined as the standard deviation of the industry-adjusted return on assets (ROA). This measure reflects the stability and sustainability of corporate profits, capturing the firm's

risk-taking from both internal and external perspectives. Higher earnings volatility signifies a greater level of corporate risk-taking. We calculate the industry-adjusted ROA as follows:

$$Adj_ROA_{i,t} = \frac{EBIT_{i,t}}{ASSET_{i,t}} - \frac{1}{X}\sum_{k=1}^{X}\frac{EBIT_{i,t}}{ASSET_{i,t}} \tag{1}$$

where $Adj_ROA_{i,t}$ represents the industry-adjusted Return on Assets for firm i in year t. $\frac{EBIT_{i,t}}{ASSET_{i,t}}$ represents ROA of firm i in year t, where EBIT denotes earnings before interest and tax, and ASSET denotes total assets. $\sum_{k=1}^{X}\frac{EBIT_{i,t}}{ASSET_{i,t}}$ represents the total ROA of all firms in the industry. Then, we adopt a five-year rolling time window to calculate the standard deviation of the adjusted ROA. A higher value of *S_Risk-taking* suggests greater financial risk for the firm.

3.2.3. Moderator

Supplier Dependence

Following Hui et al. (2019) and Dhaliwal et al. (2016), the client supplier dependence (*C_SD)* is measured by the proportion of its total annual procurement expenditure attributable to its primary supplier. Where the primary supplier is defined as the one from which the client firm makes the largest purchases. A higher *C_SD* value indicates that a client firm relies more on its largest supplier. The calculation formula is as follows:

$$C_SD = \frac{\text{Procurement amount from largest supplier}}{\text{Total procurement amount}} \times 100\% \tag{2}$$

Corporate Reputation

Following prior studies that conceptualize corporate reputation as a multi-dimensional construct shaped by stakeholders' evaluations of firm performance (Roberts and Dowling, 2002), we select twelve indicators reflecting stakeholder perspectives and use principal component factor analysis (PCFA) to reduce the dimensionality of the data and construct the client reputation indicator (*C_Reputation*).[3]

Specifically, the twelve original metrics that influence shareholders' evaluation of corporate reputation include: (1) total asset ranking, (2) revenue ranking, (3) net profit ranking, (4) market value ranking, (5) debt-to-asset ratio, (6) long-term debt ratio, (7) earnings per share, (8) proportion of independent directors, (9) earnings before tax per share, (10) sustainable asset growth rate, (11) current ratio, and (12) audit by a big four accounting firm dummy. We first adopt the PCFA methodology to select factors. As shown in Panel A of Table 1, there are four factors' eigenvalues are greater than 1, effectively explaining 61.72 % of the normalized variance of the twelve original metrics. We therefore retain these four factors and compute the client reputation indicator (*C_Reputation*) as a weighted average, where the weights are given by each factor's proportion of explanatory power relative to the total variance explained. The calculation is formulated as follows:

$$C_Reputation = \left(\frac{0.303}{0.617}\right) \times \text{Factor1} + \left(\frac{0.143}{0.617}\right) \times \text{Factor2} + \left(\frac{0.088}{0.617}\right) \times \text{Factor3} + \left(\frac{0.084}{0.617}\right) \times \text{Factor4}. \tag{3}$$

Investor Confidence

As proposed by the theory of Shleifer and Vishny (SV) and Rhodes-Kropf and Viswanathan (RKV), the market-to-book ratio of a firm can be decomposed into two distinct components: the ratio of market value to true value (M/V), and the ratio of true value to book value (V/B). Consequently, any measure of market value can be expressed using the following algebraic identity to decompose the market-to-book ratio:

$$m - b = (\mathrm{m} - \mathrm{v}) + (\mathrm{v} - \mathrm{b}) \tag{4}$$

Where m represents the market value, b denotes the book value, and v signifies a measure of the fundamental or "true" value, with all values expressed in logarithmic form.[4] If markets are prone to errors in estimating a firm's fundamental value, the difference between price and true value, $(m-v)$, captures the degree of misvaluation, which reflects investor confidence or sentiment towards the firm. Specifically, when the market price deviates from the true value, the value of $(m-v)$ will be positive, signifying overvaluation heightened investor confidence.

We construct our investor confidence indicator (*C_Investor_confidence*) following the method developed by Rhodes–Kropf et al. (2005) and Huang et al. (2015). We build a regression model to decompose the firm's market value:

$$Q_{i,t} = \alpha_0 + \alpha_1 size_{i,t} + \alpha_2 lev_{i,t} + \alpha_3 roa_{i,t} + \varepsilon_{i,t} \tag{5}$$

In this model, $Q_{i,t}$ represents the average market value of equity (market capitalization) for firm i in year t, calculated by multiplying the average share price by the number of outstanding shares. Following common practice, we use three key measures of firm performance to estimate the intrinsic book value($Q^f_{i,t}$)[5]: size, leverage, and return on assets, denoted as $size_{i,t}$, $lev_{i,t}$ and $roa_{i,t}$. We use $\varepsilon_{i,t}$ as the investor confidence indicator (*C_investor confidence*), reflecting the difference between the market value $Q_{i,t}$ and the estimated true intrinsic book value $Q^f_{i,t}$. The larger the value, the more optimistic investors become about the enterprise's development prospects and the stronger their subjective confidence grows.

3.2.4. Control variables

To account for other factors that may influence firms' ESG performance, we include a set of control variables covering financial performance, corporate governance, operational efficiency, and investor structure. Specifically, the financial performance variables include firm size, leverage, return on assets, growth, cash flow, and Tobin's Q. Corporate governance variables include CEO duality, board independence, ownership occupation, and whether the firm is audited by a Big 4 accounting firm. Operational efficiency controls include total asset turnover, inventory ratio, fixed asset ratio, and a management efficiency index. Investor structure variables include institutional ownership, top ten shareholders' concentration, and equity balance. Table 2 provides detailed definitions of all variables.

3.3. Model design

We examine the impact of supplier risk-taking on client ESG performance using the following model:

$$C_ESG_{it} = \alpha + \beta \cdot S_Risk\text{-}taking_{it} + \omega \cdot Others_{it} + Years_t + Industry_{it} + firm_i + \varepsilon_{it} \tag{6}$$

where C_ESG_{it} represents the ESG performance of client firm i in year t; $S_Risk\text{-}taking_{it}$ is the core independent variable capturing the suppliers' risk-taking behavior; $\mathrm{Others_{it}}$ denotes a vector of control variables. $\mathrm{Years_t}$, $\mathrm{Industry_{it}}$, and $\mathrm{firm_i}$ represent year, industry, and firm fixed effects, respectively. ε_{it} is the error term capturing unobserved firm-year

[3] We replace the current-period reputation measure with the average value over the previous two periods (t−1 and t−2). This approach helps mitigate the impact of reverse causality and, to some extent, alleviates the endogeneity concern regarding reputation.

[4] During our discussion, we adopt lower-case letters to represent values expressed in logarithmic form, while upper-case letters are used to denote the same values expressed in their standard units.

[5] $Q^f_{i,t} = \widehat{\alpha_0} + \widehat{\alpha_1} size_{i,t} + \widehat{\alpha_2} lev_{i,t} + \widehat{\alpha_3} roa_{i,t}$

Table 1
Factor selection.

Panel A: Factor analysis				Panel B Factor loadings				
Factor	Eigenvalue	Proportion	Cumulative	Variable	Facto1	Facto2	Facto3	Facto4
Factor1	3.630	0.303	0.303	Variable1	0.942	0.074	−0.056	0.001
Factor2	1.720	0.143	0.446	Variable2	0.909	0.089	−0.091	−0.016
Factor3	1.051	0.088	0.533	Variable3	0.671	0.502	−0.078	−0.033
Factor4	1.005	0.084	0.617	Variable4	0.923	0.105	−0.051	−0.003
Factor5	0.998	0.083	0.700	Variable5	0.456	−0.722	0.072	−0.071
Factor6	0.888	0.074	0.774	Variable6	0.384	−0.487	0.222	0.134
Factor7	0.836	0.070	0.844	Variable7	0.003	0.004	−0.294	−0.405
Factor8	0.810	0.068	0.912	Variable8	0.012	0.045	0.822	0.027
Factor9	0.496	0.041	0.953	Variable9	0.152	0.548	0.161	−0.192
Factor10	0.330	0.028	0.980	Variable10	0.008	0.194	−0.170	0.878
Factor11	0.174	0.015	0.995	Variable11	−0.344	0.574	0.134	−0.070
Factor12	0.062	0.005	1.000	Variable12	0.343	0.129	0.378	0.052

factors. Standard errors are clustered at the firm level.

Then, following prior research (Du et al., 2020), we adopt a partially linear functional-coefficient panel-data model (also known as a partially linear varying coefficient model) to test the moderating role of corporate reputation, investor confidence, and supplier dependence. The specification is outlined below:

$$C_ESG_{it} = c + \beta(Z_{it}) \cdot S_Risk\text{-}taking_{it} + \omega \cdot Others_{it} + Years_t + Industry_{it} + firm_i + \varepsilon_{it} \tag{7}$$

where Z_{it} represents a specific feature of the client firms that we are interested in (i.e. indicators of corporate reputation, investor confidence and supplier dependence), and the effect of supplier risk-taking on client ESG performance is modeled as an unknown function $\beta(\cdot)$ of Z_{it}. This model accommodates nonlinear relationships in supplier risk-taking, enabling the effects of its covariate on the dependent variable to vary as a function of a specific variable (Z) through a nonparametric approach, while preserving linear relationships for other regressors. This approach provides substantial advantages in capturing nonlinear dynamics (Cai et al., 2019; Feng et al., 2017).[6] While the results from this model should be interpreted with caution and seen as indicative rather than definitive, they offer more valuable insights into key issues than conventional linear models when examining moderating effects.

Moreover, following Cai et al. (2000), Cai (2007), Kreiss et al. (2008), we test the appropriateness of the functional form assumptions by calculating a test statistic and using the wild bootstrap method to obtain its sample distribution. This procedure is conducted separately for each moderating variable to assess the robustness of model (7). The results show that all partially linear functional-coefficient moderating effect models in this study significantly outperform the conventional linear interaction-term model. Related details and results are provided in Appendix Table 1.

4. Results

4.1. Descriptive statistics

Table 3 reports the descriptive statistics for main variables used in the study. The average ESG performance of client firms (*C_ESG*) are 4.010, with a standard deviation of 1.005, and ranges from 1.25 (1 %) to 6.00 (99 %), indicating substantial variation across firms. The mean of supplier risk-taking (*S_Risk-taking*) is 0.025, with a low standard deviation of 0.020, suggesting that most suppliers engage in relatively stable operational activities, although some exhibit significantly higher risk levels. The sample firms exhibit substantial heterogeneity across financial, governance, and structural dimensions, and there are no extreme values between the 1 % and 99 % quantiles of all variables.

4.2. Baseline regression

Table 4 reports the baseline regression results. columns (1) to (5) progressively include control variables from four dimensions: financial performance, corporate governance, operational efficiency, and investor structure. Across all model specifications, the coefficient of the key explanatory variable, *S_Risk-taking*, remains significantly positive ($P < 0.01$), suggesting that higher levels of supplier risk-taking are consistently associated with enhanced ESG performance in client firms. Using the results from column (5) in Table 4 as an example, one standard deviation increases in supplier risk-taking (0.020) is associated with an improvement of approximately 0.075 in the ESG performance of client firms ($3.742 \times 0.020 \approx 0.075$), suggesting a potential increase of approximately 1.87 % ($0.075/4.010 \times 100\ \% \approx 1.87\ \%$) in the average ESG rating of client firms.

4.3. Moderating effects

To systematically examine how supplier risk affects client firms' ESG performance under varying stakeholder conditions, this study adopts a partially linear functional-coefficient panel-data model. We identify moderating mechanisms from three key stakeholder perspectives: investors (measured by client firms' investor confidence), the public (reflected in corporate reputation), and suppliers (captured by client firms' dependence on suppliers).

Fig. 1 and Table 5 present the moderating effect of supplier dependence.[7] The results show that as client firms' supplier dependence increases, the coefficient on supplier risk-taking in explaining client ESG performance rises incrementally, exhibiting a monotonically increasing trend. At low dependence quantiles, this effect is negative and insignificant, indicating that when firms have ample alternative suppliers, they are less sensitive to risk-taking by their primary supplier and

[6] Following Libois and Verardi (2013), we use the series method, specifically the B-splines technique to fit model (7). More technical details on B-splines can be found in Newson (2001).

[7] Due to the continuity of the sample data, accurately estimating coefficients at individual quantile points is challenging. We therefore average the estimation results around each target quantile to obtain the coefficients and t-statistics reported in Table 5. For example, for the 0.1 quantile, we average the results from the 0.09 to 0.11 quantiles to generate the smoothed values shown in Column (1) of Table 5. All tables in this section are constructed using this approach.

Table 2
Variable definitions.

Variable	Definition	Sources
Client's ESG		
C_ESG	The ESG performance of clients. Including three dimensions: Environmental (E), Social (S), and Corporate Governance (G). It captures how well a company manages environmental impact, social responsibilities, and governance practices to support long-term sustainability.	Sino-Securities Index ESG Ratings
Supplier's risk-taking		
S_Risk-taking	The risk-taking of suppliers. Measured by the standard deviation of industry-adjusted return on assets (ROA). The higher its value, the more volatile the supplier's earnings are, implying a higher level of risk-taking.	CSMAR and calculated by authors
Moderators		
Supplier dependence	The proportion of annual procurement sourced from its largest supplier (%).	CSMAR and calculated by authors
Corporate reputation	A composite indicator reflecting stakeholders' evaluations of firm performance, constructed using principal component analysis. A higher value indicates a better reputation.	CSMAR and calculated by authors
Investor confidence	A composite indicator developed by Rhodes–Kropf et al. (2005) and Huang et al. (2015) to measure investor confidence. The higher the value, the higher the investor confidence.	CSMAR and calculated by authors
Financial performance		
Size	The logarithm of total assets.	CSMAR and calculated by authors
Leverage	The ratio of total liability to total assets.	CSMAR and calculated by authors
ROA	The return to assets ratio.	CSMAR and calculated by authors
Growth	The ratio of (current period's operating income - previous operating income) to previous period's operating income.	CSMAR and calculated by authors
Cashflow	The ratio of net cash flows from operating activities to current liabilities.	CSMAR and calculated by authors
TobinQ	The ratio of (market value of total assets + book value of total liabilities)/to book value of total assets.	CSMAR and calculated by authors
Corporate governance		
Dual	A dummy variable that takes the value of 1 means that the CEO and Chairman are the same person, while a value of 0 means that they are not.	CSMAR and calculated by authors
Independent	Percentage of independent directors. It is measured as the ratio of the number of independent directors to the total number of board members.	CSMAR and calculated by authors
Occupy	Occupation of funds by major shareholders. Measured by the ratio of major shareholder loan balances to total assets.	CSMAR and calculated by authors
Big4	A dummy variable with a value of 1 indicates that the company is audited by a Big 4 accounting firm, and a value of 0 indicates otherwise.	CSMAR and calculated by authors
Operational efficiency		
ATO	Total asset turnover. Measured by the ratio of net sales to average total assets. The larger the value, the more efficiently the firm uses its total assets to generate revenue.	CSMAR and calculated by authors
Inventory	The ratio of inventory balance to total assets. Higher values of it imply that the company is investing a large amount of capital in inventory, which may lead to lower liquidity.	CSMAR and calculated by authors
Fixed	The ratio of net fixed assets to total assets. A higher value of it reflects a company's investment in long-term assets.	CSMAR and calculated by authors
Management efficiency	An index that measures the degree of cost-effectiveness. It is measured as the ratio of administrative expenses to operating income. The higher the value, the less efficient it is.	CSMAR and calculated by authors
Investor Structure		
INST	Institutional investor holdings. Measured by institutional shareholding to total outstanding shares.	CSMAR and calculated by authors
Top10	Top ten shareholders' shareholdings. Measured by Top ten shareholders' shareholdings to total outstanding shares.	CSMAR and calculated by authors
Balance	Degree of Equity Balance. It is measured using the percentage of shares held by the top five shareholders minus the percentage of shares held by the first largest shareholder.	CSMAR and calculated by authors

supplier risk does not effectively trigger an ESG response. By contrast, once dependence reaches 0.3 quantile to high quantiles, the coefficient becomes significantly positive, suggesting that high dependence heightens clients' concern for supply chain stability and prompts them to manage risk more proactively through ESG practices. These findings support Hypothesis 2.

Fig. 2 and Table 6 illustrate the moderating effect of corporate reputation. As shown in column (1) of Table 6, at low quantiles, supplier risk has a significant negative effect on ESG performance, indicating that low-reputation firms—often with limited media exposure—face weaker stakeholder pressure and may lack the capacity for proactive ESG responses. These firms may even reduce ESG efforts due to risk shifting or resource constraints. As reputation increases, the effect turns positive and rises sharply in the upper quantiles, suggesting that high-reputation firms, driven by stronger image-preservation motives, are more sensitive and responsive to upstream risks. This reflects a cost–benefit trade-off: while low-reputation firms prioritize short-term operational efficiency, high-reputation firms invest in ESG as a strategic buffer against external uncertainties. These findings support Hypothesis 3 .[8]

Fig. 3 and Table 7 illustrate the moderating effect of investor confidence on the relationship between supplier risk-taking and client firms' ESG performance. The results reveal a monotonic decreasing trend: when investor confidence is low, supplier risk has a significantly positive effect on ESG performance, suggesting that firms strengthen ESG efforts to signal stability and reliability amid market skepticism. However, as shown in column (5) of Table 7, as investor confidence increases, the positive effect diminishes and eventually approaches zero. This pattern implies that firms enjoying high investor confidence face less external pressure and are less compelled to use ESG as a signaling mechanism, thereby weakening their ESG response to supplier-related risks. These

[8] We also assign this indicator into ten levels based on decile rankings to construct the reputation ranking indicator, a common practice in existing literature, and use it as a moderating variable. The estimation results remain robust.

Table 3
Descriptive statistics.

Variable	Mean	Std. dev.	1 %	Median	99 %	Observations
Client's ESG						
C_ESG	4.010	1.005	1.250	4.000	6.000	2800
Supplier's risk-taking						
S_Risk-taking	0.025	0.020	0.004	0.018	0.106	2800
Moderator						
Supplier dependence	17.430	15.608	1.780	12.510	77.440	2647
Corporate reputation	0.071	0.546	−0.885	0.020	1.514	2399
Investor confidence	−0.106	0.862	−1.610	−0.243	3.316	2761
Financial performance						
Size	22.471	1.422	20.059	22.238	26.738	2800
Leverage	0.463	0.203	0.069	0.462	0.888	2800
ROA	0.037	0.060	−0.187	0.037	0.172	2800
Growth	0.161	0.393	−0.505	0.104	1.902	2800
Cashflow	0.045	0.066	−0.133	0.045	0.215	2800
TobinQ	1.823	1.087	0.849	1.456	6.598	2800
Corporate governance						
Dual	0.221	0.415	0.000	0.000	1.000	2800
Independent	0.369	0.050	0.300	0.333	0.556	2800
Occupy	0.017	0.028	0.000	0.007	0.130	2800
Big4	0.063	0.243	0.000	0.000	1.000	2800
Operational efficiency						
ATO	0.640	0.400	0.104	0.558	2.000	2800
Inventory	0.132	0.132	0.000	0.102	0.717	2800
Fixed	0.230	0.174	0.004	0.188	0.703	2800
Management efficiency	0.084	0.067	0.011	0.068	0.359	2800
Investor Structure						
INST	0.466	0.243	0.005	0.492	0.935	2800
Top10	0.584	0.153	0.237	0.590	0.924	2800
Balance	0.750	0.659	0.024	0.574	2.912	2800

findings support Hypothesis 4.

4.4. Robustness and endogeneity tests

4.4.1. Robustness tests

Table 8 reports robustness checks on the baseline estimates. In column (1), to rule out the possibility that the positive link between supplier risk-taking and client ESG performance is driven by supplier financial violations (Hu et al., 2024),[9] we construct a panel vector autoregressive (Panel VAR) model to isolate the portion of ROA volatility unrelated to financial violations and use it to generate a revised risk-taking measure, denoted as σ (ROA) excluding financial violations. The results show a significant positive relationship, indicating that even after removing the effects of supplier violations on financial volatility, client firms' ESG performance still improves. This supports our argument that, unlike passive, post-event ESG adjustments triggered by supplier violations, clients treat ESG as a proactive strategy for risk prevention and absorption in response to uncertainties arising from upstream suppliers' risk-taking. In columns (2) and (3) of Table 8, we replace the original supplier risk-taking indicator with the Z-score and the standard deviation of ROE, respectively.[10]

In column (4), following Priem and Gabellone (2024), we change our ESG rating indicator into ESG score from the Sino-Securities Index ESG Ratings dataset. Column (5) employs Wind ESG rating as the dependent variable to replicate the baseline regressions. Wind ESG evaluates multiple aspects of ESG investment, practices, disclosure, and regulations, providing 27 subjects and over 300 individual elements related to three specific ESG areas. Like the Sino-Securities ESG data, the Wind ESG dataset covers almost all listed companies in China and is highly tailored to the country, accurately reflecting its realities (Wu et al., 2024). To further increase the generalizability and credibility of the result, we also incorporate additional international ESG data sources, including Bloomberg, Refinitiv and Morgan Stanley Capital International (MSCI) for robustness checks in columns (6)–(8), the results are all qualitatively unchanged.[11]

Next, the baseline model is re-estimated using unconditional quantile regression at the median of the dependent variable, a panel logit model and the System GMM approach in columns (9)–(11). All other control variables from the baseline model are included in each specification.

Then, to exclude the impact of the global COVID-19 pandemic-induced recession on both supply chains and corporate decision-making, we reconduct the baseline regression using sample from 2009 to 2019. As shown in Column (12), the coefficient of *S_Risk-taking* also remains positive and significant. In summary, all the results remain consistent with our baseline findings, confirming the robustness of H1.

4.4.2. Instrumental variable (IV) approach

While high-ESG-performing firms generally emphasize sustainability and risk mitigation (Albuquerque et al., 2019), their strong ESG profile may also increase their tolerance for upstream operational risk, raising concerns of potential reverse causality. To address this, we employ suppliers' inventory turnover (*Turnover_G*) as an instrumental variable. Inventory turnover is widely regarded as an important indicator of a firm's operational efficiency and risk exposure. A lower inventory turnover typically indicates excessive inventory holdings, leading to capital tie-up and increased liquidity risk (Aktas et al., 2015), and may

[9] As a representative of supplier misconduct, financial violations are believed to be transmitted through the supply chain, prompting client firms to passively improve their ESG performance (Hu et al., 2024).

[10] Although not reported here, we also calculate the standard deviation of ROA using a three-year rolling window and use it as our supplier risk-taking indicator to check robustness. The results remain consistent.

[11] Due to the limited coverage of international ESG databases for Chinese listed firms, the matched sample size is substantially reduced. This coverage gap may arise from differences in firm participation, disclosure transparency, data availability, and calculation methodologies. Under such conditions, including firm fixed effects would excessively consume degrees of freedom, weaken statistical power, and risk unstable estimates. Therefore, we control only for industry and year fixed effects in these regressions.

Table 4
The impact of supplier risk-taking on client ESG performance.

Dependent variable					
	(1)	(2)	(3)	(4)	(5)
	C_ESG	*C_ESG*	*C_ESG*	*C_ESG*	*C_ESG*
S_Risk-taking	3.800** (2.501)	3.755*** (2.587)	3.614** (2.517)	3.937*** (2.827)	3.742*** (2.650)
Financial performance					
Size		0.363***	0.375***	0.446***	0.478***
		(2.972)	(3.225)	(3.840)	(3.856)
Leverage		−1.180***	−1.164***	−1.528***	−1.640***
		(-3.859)	(-4.034)	(-5.352)	(-5.551)
ROA		0.011	−0.108	−0.401	−0.497
		(0.019)	(-0.191)	(-0.603)	(-0.728)
Growth		−0.099*	−0.095*	−0.219***	−0.221***
		(-1.764)	(-1.736)	(-3.783)	(-3.788)
Cashflow		0.315	0.257	0.424	0.437
		(0.908)	(0.748)	(1.247)	(1.256)
TobinQ		−0.013	−0.012	−0.010	−0.005
		(-0.392)	(-0.364)	(-0.341)	(-0.145)
Corporate governance					
Dual			−0.023	0.028	0.017
			(-0.225)	(0.271)	(0.164)
Independent			0.755	0.697	0.765
			(1.151)	(1.003)	(1.099)
Occupy			−1.373	−1.006	−0.996
			(-1.001)	(-0.789)	(-0.787)
Big4			−0.756**	−0.943**	−0.914**
			(-1.975)	(-2.350)	(-2.281)
Operational efficiency					
ATO				0.378*	0.407*
				(1.701)	(1.812)
Inventory				2.057***	2.059***
				(2.795)	(2.802)
Fixed				−0.474	−0.486
				(-1.215)	(-1.237)
Management efficiency				−2.495***	−2.495***
				(-3.649)	(-3.607)
Investor Structure					
INST					−0.453
					(-0.915)
Top10					0.420
					(0.716)
Balance					−0.138
					(-1.176)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	2934	2879	2867	2801	2800
R^2	0.873	0.873	0.874	0.876	0.877

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

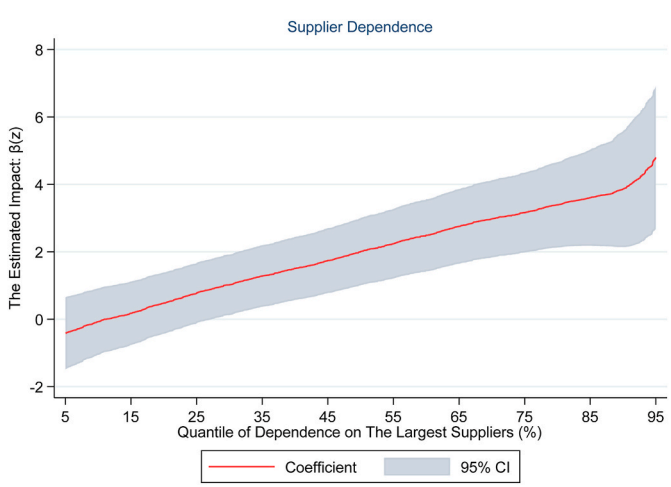

Fig. 1. The moderating role of supplier dependence.

Table 5
The moderating role of supplier dependence.

Dependent variable: C_ESG					
	(1)	(2)	(3)	(4)	(5)
Quantile of moderator	*0.1*	*0.3*	*0.5*	*0.7*	*0.9*
S_Risk-taking	−0.069	1.028**	2.000***	2.974***	3.868***
	(-0.122)	(2.056)	(3.523)	(4.443)	(4.079)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	2647	2647	2647	2647	2647

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets.

also reflect inaccurate demand forecasting, thereby reducing the stability of profitability (Eroglu and Hofer, 2011). This factor can influence a firm's willingness and capacity for strategic risk-taking (e.g., through financing constraints or adjustments to business strategy), thus satisfying the relevance condition. On the other hand, the exclusion restriction assumes that inventory turnover affects client ESG performance only through its influence on supplier risk-taking. This assumption is

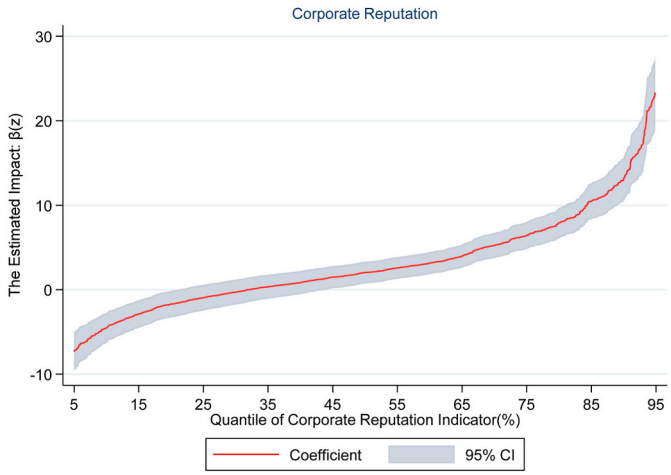

Fig. 2. The moderating role of client corporate reputation.

Table 6
The moderating role of client corporate reputation.

Dependent variable: C_ESG					
	(1)	(2)	(3)	(4)	(5)
Quantile of moderator	*0.1*	*0.3*	*0.5*	*0.7*	*0.9*
S_Risk-taking	−4.480***	−0.281	2.017***	5.246***	13.300***
	(-5.262)	(-0.395)	(2.964)	(6.870)	(11.635)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	2069	2069	2069	2069	2069

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets.

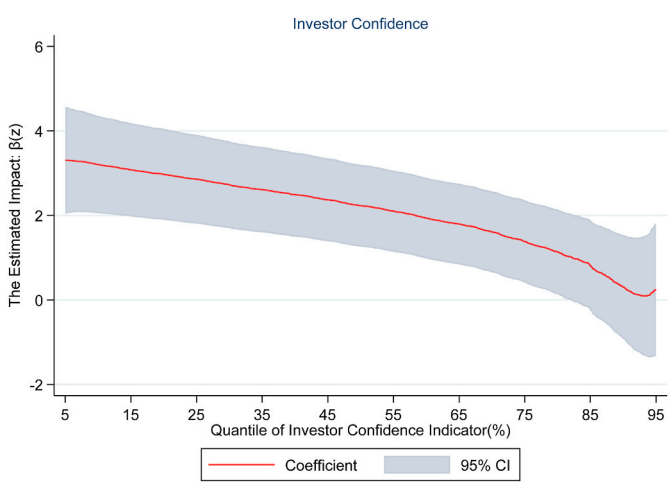

Fig. 3. The moderating role of client investor confidence.

reasonable because inventory turnover primarily reflects suppliers' internal operational efficiency, which is not institutionally linked to client ESG activities. Moreover, the main drivers of client ESG performance—such as governance policies and stakeholder relations (Albuquerque et al., 2019; Grewal et al., 2016)—are largely unrelated to the short-term inventory management practices of upstream suppliers.

Table 9 shows that the instrument passes the under-identification ($p < 0.05$), weak identification (57.707 > 16.38), and weak-instrument ($p < 0.01$) tests, confirming its strength and validity. Column (1) of Table 9 shows that *Turnover_G* is significantly and negatively related to supplier

Table 7
The moderating role of client investor confidence.

Dependent variable: C_ESG					
	(1)	(2)	(3)	(4)	(5)
Quantile of moderator	*0.1*	*0.3*	*0.5*	*0.7*	*0.9*
S_Risk-taking	3.205***	2.722***	2.231***	1.612***	0.303
	(6.128)	(5.841)	(5.160)	(3.875)	(0.564)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	2761	2761	2761	2761	2761

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets.

risk-taking, while Column (2) indicates that, after correcting for endogeneity, supplier risk-taking remains significantly and positively associated with client ESG performance ($\beta = 24.8926$, $p < 0.05$).

4.4.3. PSM method

To address potential selection bias from the non-random matching between clients and high-risk suppliers, we use propensity score matching (PSM) with kernel matching. Clients whose primary suppliers fall above the 90th percentile in risk-taking form the treatment group. The kernel method assigns weights to all control units based on their proximity in propensity scores, resulting in smoother and more efficient estimates.[12] Table 10 reports the regression results based on the matched sample. columns (1) to (5) show that after PSM method, all the coefficients of supplier risk-taking remain consistent with the baseline findings, supporting the robustness of the main results.

5. Extended analyses

5.1. Different types of supplier risk-taking behaviors

Prior research suggests that outward foreign direct investment (OFDI), R&D expenditure, and asset financialization are often regarded as indicators of a firm's propensity to undertake higher levels of risk in pursuit of long-term strategic returns (Li, 2011; Nielsen and Nielsen, 2011; Bargeron et al., 2010). Although these "high-risk, high-return" behaviors may not necessarily lead to operational or financial distress for the supplier, they introduce supply chain uncertainty for client firms.

First, OFDI often requires suppliers not only to allocate part of the funds and products needed for domestic operations to overseas markets, but also to adjust their product structure and business strategies according to the characteristics of the target regions, while seeking new client firms based on local demand. These actions lead to adjustments in domestic production and the restructuring of the supply chain, thereby increasing the uncertainty faced by local client firms. Second, increased R&D investment may divert funds away from the supplier's production of existing products, thereby altering the supply of raw materials and intermediate goods, which disrupts the client firm's production plans. Additionally, changes in the quality of raw materials and intermediate goods resulting from supplier product innovation may require adjustments in the client firm's existing production technologies and processes, thereby increasing uncertainty in production processes and costs (Wu et al., 2022). As last, although higher financialization, i.e. the increase of financial asset holdings (such as securities), may increase

[12] Balance tests confirm that the matching procedure substantially improved covariate balance: mean bias decreased from 7.2 % to 1.0 %, and pseudo-R^2 dropped from 0.015 to near zero. Most standardized biases fall below the 5 % threshold, and post-matching t-tests show no significant differences between treated and control groups. The ATT estimate indicates that firms with high-risk suppliers have slightly higher ESG scores compared to those with lower-risk suppliers.

Table 8
Robustness tests.

Dependent variable: C_ESG	**Different indicator of supplier's risk-taking**			**Different indicator of client ESG**				
	(1) *σ(ROA) excluding financial violation*	(2) *Z-score*	(3) *σ(ROE)*	(4) *Sino-Securities ESG Score*	(5) *Wind ESG*	(6) *Bloomberg ESG*	(7) *Refinitiv ESG*	(8) *MSCI ESG*
S_Risk-taking	2.197***	−0.092**	0.917**	19.539**	2.933*	23.791*	1.506***	9.102*
	(2.914)	(-2.276)	(2.131)	(2.223)	(1.867)	(1.892)	(3.459)	(1.739)
Control variables	Yes	Yes	Yes		Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes			
Observations	2292	2765	2657	Yes	2800	1036	367	176
R^2	0.876	0.879	0.878	2772	0.750	0.750	0.752	0.860
	Different econometric methodologies			**Different sample period**				
	(9)	(10)	(11)	(12)				
Dependent variable: *C_ESG*	*Unconditional quantile regression 50th*	*Panel logit*	*System GMM*	*Exclude the epidemic period*				
S_Risk-taking	6.180***	13.122***	4.513**	3.236**				
	(3.106)	(2.840)	(2.063)	(1.979)				
Control variables	Yes	Yes	Yes	Yes				
Industry FE	Yes	Yes	Yes	Yes				
Year FE	Yes	Yes	Yes	Yes				
Firm FE	Yes	Yes	Yes	Yes				
Observations	2800	2778	2459	2137				
R^2	0.817			0.893				
AR (1)/AR (2)			0.000/ 0.119					
Hansen J			0.102					

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

Table 9
The regression results using instrumental variable (IV) approach.

Dependent variable: C_ESG	IV	
	(1) S_Risk-taking	(2) *C_ESG*
S_Risk-taking		24.8926**
		(2.1586)
Turnover_G	−0.0085***	
	(-2.9911)	
Control variables	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Firm FE	Yes	Yes
Observations	2798	2798
Under-identification test	Chi-square statistics [p values]	
Kleibergen-Paap rk LM statistic	4.668 ** [0.0307]	
Weak identification test		
Cragg-Donald Wald F statistic	57.707	
Stock-Yogo weak ID test critical values	16.38 (10 %)	
Weak-instrument test	Chi-square statistics [p values]	
Anderson-Rubin Wald test	8.89 *** [0.0029]	

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

revenue diversification and reduce the negative impact of core business disruptions on profitability, the increased funds required to purchase financial assets may crowd out the supplier's capital for actual production, weakening the stability of product supply and, in turn, increasing the production uncertainty for client firms (Ma et al., 2022).

Therefore, we further conduct an analysis to investigate the impact of these three specific upstream risk-taking behaviors on client ESG performance. Specifically, (1) the OFDI indicator is a dummy variable equal to 1 if the company has OFDI and 0 otherwise; (2) the R&D indicator, measured as the ratio of R&D expenditure to profits, reflects the share of profits devoted to innovation; and (3) the financialization indicator, defined as the change in the ratio of available-for-sale financial assets to total assets, captures the shift toward liquid financial investments. The results of Table 11 show that typical risk-taking behaviors of supplier firms significantly enhance the ESG performance of client firms. This finding reinforces the argument that, when exposed to upstream risk activities, client firms strategically strengthen their ESG practices to signal responsible governance and thus provide empirical support for H1.

5.2. The impact of supplier risk-taking on components of client ESG performance

Further, we decompose the ESG composite score into three components, E (environmental), S (social), and G (governance). Table 12 illustrates the impact of supplier risk-taking on three components of client ESG score.

Column (1) shows that supplier risk-taking has no significant effect on the environmental (E) score of client firms, suggesting that firms do not substantially increase environmental investment in response to upstream potential risks. This may be due to the high capital intensity and technical complexity of environmental initiatives, which limit their feasibility as short-term responses.

In contrast, columns (2) and (3) indicate that supplier risk-taking significantly improves client firms' social (S) and governance (G) performance. One possible explanation is that compared to environmental efforts, social and governance responses are less resource-intensive and more adaptable in the short term (Distelhorst et al., 2017; Kalkanci and Plambeck, 2020b), making them preferred strategic adjustments under rising supply chain uncertainty.

5.3. The moderating role of corporate digitalization

In recent years, corporate digitalization has become a defining trend in global supply chain management, fundamentally reshaping operational models and enhancing firms' capacity to address external uncertainty and multifaceted risks (Ivanov, 2024). From a stakeholder perspective, digitalization enables faster alignment of ESG initiatives

Table 10
PSM estimates of supplier risk-taking impact on client ESG performance.

Dependent variable					
	(1)	(2)	(3)	(4)	(5)
	C_ESG	*C_ESG*	*C_ESG*	*C_ESG*	*C_ESG*
S_Risk-taking	3.4177*	3.3608**	3.1118*	3.4425**	3.3392**
	(1.9558)	(2.0416)	(1.9455)	(2.2045)	(2.1227)
Financial performance					
Size		0.4056***	0.4170***	0.4659***	0.5107***
		(2.9262)	(2.9614)	(3.2955)	(3.4530)
Leverage		−1.0311***	−0.9448***	−1.2871***	−1.4121***
		(-3.1897)	(-2.9506)	(-3.9822)	(-4.2251)
ROA		0.6655	0.5586	−0.3481	−0.4551
		(1.1271)	(0.9663)	(-0.5405)	(-0.6768)
Growth		−0.1282	−0.1189	−0.1842**	−0.1891**
		(-1.6122)	(-1.5758)	(-2.4377)	(-2.5318)
Cashflow		0.2033	0.1274	0.2591	0.2597
		(0.4838)	(0.3146)	(0.6698)	(0.6650)
TobinQ		−0.0092	−0.0039	−0.0021	0.0021
		(-0.2710)	(-0.1184)	(-0.0663)	(0.0634)
Corporate governance					
Dual			−0.0188	0.0167	0.0132
			(-0.1705)	(0.1540)	(0.1204)
Independent			1.3202**	1.2619*	1.3088*
			(2.0478)	(1.8681)	(1.9210)
Occupy			−1.8278	−1.6471	−1.5031
			(-1.0738)	(-0.8626)	(-0.7817)
Big4			−1.0656**	−1.0341**	−1.0053**
			(-2.5302)	(-2.3161)	(-2.2625)
Operational efficiency					
ATO				0.4133	0.4515
				(1.4310)	(1.5569)
Inventory				2.4139***	2.4411***
				(2.8160)	(2.8467)
Fixed				−0.5383	−0.5335
				(-1.3534)	(-1.3231)
Management efficiency				−2.1463***	−2.1021***
				(-2.9159)	(-2.8316)
Investor Structure					
INST					−0.3607
					(-0.7017)
Top10					0.4634
					(0.6715)
Balance					−0.1646
					(-1.2442)
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	2660	2660	2660	2660	2660
R^2	0.8641	0.8641	0.8713	0.8766	0.8772

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

Table 11
The impact of different supplier risk-taking behaviors on client ESG performance.

Dependent variable				
	(1)	(2)	(3)	(4)
	C_ESG	*C_ESG*	*C_ESG*	*C_ESG*
OFDI	0.141**			0.152**
	(2.204)			(2.088)
R&D		0.001**		0.001**
		(2.490)		(2.422)
Financialization			1.411*	1.606**
			(1.907)	(1.983)
Control variables	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	2800	2745	2437	2434
R^2	0.877	0.876	0.880	0.881

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

Table 12
The impact of supplier risk-taking on components of client ESG performance.

Dependent variable	(1)	(2)	(3)
	Environmental	*Social*	*Governance*
S_Risk-taking	0.378	2.785*	4.587**
	(0.233)	(1.723)	(1.975)
Control variables	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	2800	2800	2800
R^2	0.900	0.866	0.858

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets and are calculated using firm-level clustered standard errors.

with stakeholder expectations by leveraging real-time, reliable data for timely and targeted responses, thereby facilitating preventive governance and value preservation before risks materialize. It strengthens internal and cross-organizational coordination, allowing ESG measures to be efficiently implemented across the supply chain—for example, generative AI fosters inter-firm collaboration, knowledge co-creation,

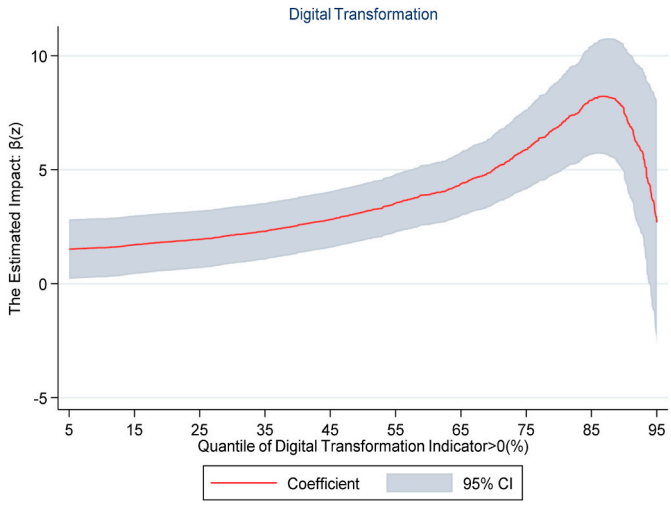

Fig. 4. The moderating role of corporate digitalization.

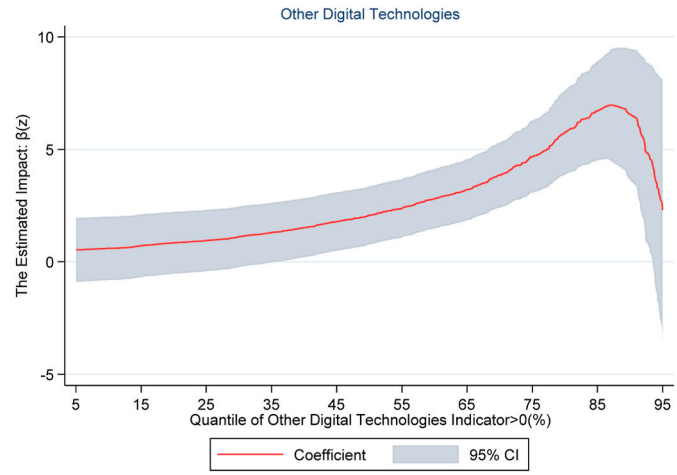

Fig. 5. The moderating role of other digital technologies.

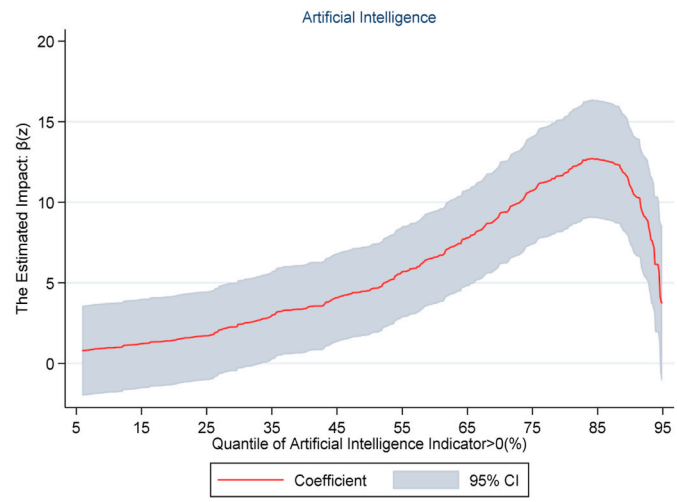

Fig. 6. The moderating role of AI.

and green innovation (Wang and Zhang, 2025), while supply chain systems and shared data platforms break down information silos, enabling real-time risk sharing and synchronized decision-making (Dubey et al., 2019). It also enhances early risk detection by integrating suppliers' operational data (e.g., production, quality, logistics, cash flow) with advanced analytics and early warning systems, enabling earlier and more accurate identification of issues from upstream risk-taking (Ivanov, 2024; Queiroz et al., 2022). Finally, through risk modeling and scenario simulation, firms can design cost-effective, targeted ESG interventions—such as enhanced audits, contractual revisions, or transparency measures—tailored to specific risk types and severities.

Building on this theoretical logic, we adopt model (7) to examine how digitalization influences the relationship between supplier risk-taking and client firms' ESG performance, where digitalization indicator (DT) is measured as the ratio of the frequency of digitalization-related terms to the total length of the annual reports. The digitalization indicator is sourced from Chen & Cao (2025). The results are presented in Fig. 4 and Table 13, respectively. We find that before the approximately 90th percentile, as the level of corporate digitalization increases, the positive impact of supplier risk-taking on client ESG continues to rise. However, it is worth noting that when corporate digitalization reaches an extreme high level, a turning point occurs, leading to a decline. A possible explanation for this phenomenon is that client firms with a high level of corporate digitalization have the capability to accurately predict the potential impact of suppliers on their supply chains using abundant information and advanced data analytics technologies. This reduces the uncertainty perceived by client firms due to supplier risk-taking, thereby decreasing their proactive response behaviors.

Next, based on feature words in different categories, we further decompose the aggregated corporate digitalization into two components: artificial intelligence (AI) and other digital technologies (ODT), to compare whether different digital technologies have heterogeneous impacts. As shown in Figs. 5 and 6, although the patterns in these two figures are identical, in the ascending phase, the curve in Fig. 6 is steeper, indicating that, compared to other digital tools, advancements in AI technology significantly enhance a firm's ability to manage ESG performance and respond to supply chain uncertainties. Second, a vertical comparison of the data from columns 1 to 5 in Panel A and columns 1 to 5 in Panel B of Table 14 shows that, at the same quantile, the coefficients in Panel A are significantly larger than those in Panel B.[13] This indicates that AI, as the most complex and advanced digital technology currently available, plays a significantly more prominent role than other digital technologies in promoting supplier risk identification, optimizing supply chain management, and enhancing corporate governance and operational efficiency.

Table 13
The moderating role of corporate digitalization.

Dependent variable:C_ESG					
	(1)	(2)	(3)	(4)	(5)
Quantile of DT	0.1	0.3	0.5	0.7	0.9
S_Risk-taking	1.575**	2.130***	3.137***	4.999***	7.481***
	(1.959)	(2.783)	(4.323)	(6.569)	(5.889)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	1877	1877	1877	1877	1877

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets.

[13] It is worth noting that, in contrast to the significantly positive coefficient in column (1) of Table 13, the coefficients in column (1) of Panel A and Panel B of Table 14 are not significant. This suggests that, at the early stages of digital transformation, focusing on the development of a single digital technology yields minimal results. Only by simultaneously investing in and applying multiple digital technologies can a firm quickly enhance its supply chain management capabilities and governance level.

Table 14
The moderating role of AI and other technologies.

Dependent variable: C_ESG					
Panel A: Artificial intelligence					
	(1)	(2)	(3)	(4)	(5)
Quantile of AI	0.1	0.3	0.5	0.7	0.9
S_Risk-taking	0.953	2.374**	4.564***	9.291***	10.987***
	(0.646)	(1.671)	(3.292)	(6.176)	(6.219)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	644	644	644	644	644
Panel B: Other digital technologies					
	(1)	(2)	(3)	(4)	(5)
Quantile of ODT	0.1	0.3	0.5	0.7	0.9
S_Risk-taking	0.595	1.113**	2.054***	3.854***	6.573***
	(0.914)	(1.778)	(3.365)	(5.595)	(4.597)
Control variables	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Observations	1852	1852	1852	1852	1852

Notes: * < 0.1, ** < 0.05, *** < 0.01. T-values are shown in brackets.

6. Conclusion and discussion

This study explores the impact of supplier risk-taking on client ESG performance, revealing significant insights into the dynamics within supply chains. Analyzing data from Chinese listed firms between 2009 and 2023, we find that supplier risk-taking has a significantly positive effect on client firms' ESG performance, indicating that firms regard ESG not merely as an internal governance mechanism, but also as a forward-looking strategic tool for managing uncertainties transmitted from upstream firms. Moreover, the study identifies key boundary conditions that shape this relationship. Specifically, higher supplier dependence, stronger corporate reputation, and lower investor confidence significantly amplify the positive impact of supplier risk-taking on ESG performance. These moderating effects suggest that as external stakeholder pressure intensifies, firms become more responsive in adjusting their ESG strategies to mitigate supplier-related risks. Notably, under certain extreme conditions—such as low supplier dependence or high client investor confidence—the positive association may weaken or even reverse, highlighting the presence of threshold effects in the ESG transmission mechanism within supply chains.

Our study contributes to both theory and practice. Theoretically, in comparison with prior research that largely interprets ESG engagement as a reactive strategy triggered by supplier misconduct or crisis events (Li and Cuervo-Cazurra, 2024; Hu et al., 2024), our study distinguishes supplier risk-taking from misconduct behavior and demonstrates that strategic risk decisions—detectable in advance through operational indicators—can proactively shape downstream ESG strategies. By doing so, our study extends stakeholder theory and signaling theory by showing that upstream suppliers' strategic risk-taking can function as a forward-looking signal that prompts downstream ESG engagement, thereby expanding the traditional view of ESG as merely a reactive response to corporate misconduct. This shifts the academic conversation toward the proactive role of ESG in risk absorption and prevention within supply chains.

Moreover, while existing literature on corporate risk-taking has primarily examined intra-firm outcomes such as innovation, performance, and financial policy (García-Granero et al., 2015; Cain and McKeon, 2016; Díez-Esteban et al., 2017), our study goes beyond this scope to inter-firm effects within supply chains, uncovering a novel transmission channel from supplier risk-taking to client ESG performance.

Finally, our results hold relevance for other emerging market economies that often face similar challenges with China, such as less mature ESG disclosure systems (Agyemang et al., 2025), weaker corporate governance, and faster-paced digital adoption. As such, the mechanisms we identify can provide actionable insights for enhancing ESG effectiveness and resilience in comparable contexts.

In practical terms, several targeted managerial implications emerge. First, client firms should treat ESG as a proactive risk management instrument, particularly when supplier dependence is high or when reputation concerns are prominent. Managers are advised to strengthen ESG practices not only to satisfy regulatory or social expectations, but also to buffer against upstream uncertainties that could threaten operational continuity and market value. Second, firms with strong reputational capital or those facing lower investor confidence should be especially vigilant in aligning ESG initiatives with supply chain risk monitoring, as stakeholder sensitivity in these contexts magnifies the consequences of supplier behaviors. Third, recognizing the threshold effects, firms should avoid over-reliance on a single supplier and diversify their supply base to prevent scenarios in which ESG alone cannot offset risks. Finally, policymakers should design ESG governance frameworks that integrate supply chain transparency, ensuring that firms are incentivized to account for upstream risks when shaping their ESG strategies.

Beyond these primary implications, our extended analysis highlights the supplementary role of corporate digitalization—particularly artificial intelligence (AI)—in strengthening clients' capacity to detect supplier risks earlier and implement targeted ESG interventions. By leveraging real-time data, advanced analytics, and collaborative platforms, firms can enhance operational resilience and supply chain transparency, thereby supporting more sustainable ESG outcomes (Ivanov, 2020; Dubey et al., 2019). In particular, AI-driven tools for knowledge management and supplier profiling provide valuable opportunities for continuous evaluation of upstream partners (Wang and Zhang, 2025), suggesting that digitalization should be integrated into ESG and risk management strategies as a complement to, rather than a substitute for, traditional practices. Moreover, at the policy level, well-designed regulatory frameworks for emerging technologies and environmental standards can further strengthen the positive role of digitalization and AI in supply chain performance and ESG governance, ensuring that technological adoption translates into more sustainable outcomes.

We acknowledge several limitations in this study, thus providing a foundation for further research. First, although our sample size is comparable to that in related supply chain studies (Geng et al., 2024; Wang et al., 2023a,b; Sun et al., 2023), the constrained institutional disclosure rules in Chinese listed firms, especially regarding the supplier information, inherently limits the number of observable supply chain relationships, and thus largely reduces the comprehensiveness of our data. Future research could benefit from incorporating broader supply chain databases to enlarge the sample coverage.

Second, corporate reputation—used here as a moderating variable—may be endogenous to client firms' long-term ESG performance. Firms with sustained high ESG performance are more likely to build strong reputations, while certain unobserved factors may simultaneously influence both variables. While this study focuses on the association rather than strict causal identification, future research could address this issue by employing dynamic panel models, instrumental variable approaches, or natural experiments to more robustly identify the causal moderating role of corporate reputation.

Third, while our theoretical framework emphasizes the proactive nature of ESG responses, our empirical design of using the annual frequency data makes it difficult to definitively distinguish proactive strategies from rapid reactions occurring within the same year. We therefore leave this as a promising avenue for future research, which could combine high-frequency (e.g., quarterly or event-level) data with event-study methods to provide deeper insights into the temporal dynamics of ESG adjustments.

CRediT authorship contribution statement

Jing Zhou: Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Tingwei Wu:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation. **Kaiwen Wu:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Formal analysis, Data curation, Conceptualization.

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Appendix

Appendix 1 Functional form test of partially linear functional-coefficient model.

In this appendix, we introduce the method for testing the stability of our moderating model. To examine whether the functional form assumptions we adopt for the moderating effects are more appropriate than those of the conventional linear interaction term model, we construct a statistic and use the wild bootstrap method to obtain the sample distribution of this statistic. Specifically, we first present the two models as follows:

$$C_ESG_{it} = \alpha + \beta_1 \cdot S_Risk\text{-}taking_{it} + \beta_2 \cdot S_Risk\text{-}taking_{it} \times Z_{it} + \omega \cdot Others_{it} + Years_t + Industry_{it} + firm_i + \varepsilon_{1,it} \tag{1a}$$

$$C_ESG_{it} = c + \beta(Z_{it}) \cdot S_Risk\text{-}taking_{it} + \omega \cdot Others_{it} + Years_t + Industry_{it} + firm_i + \varepsilon_{2,it} \tag{2a}$$

Model 1 is a linear interaction term panel data model, while Model 2 is a partially linear functional-coefficient panel data model.

Second, we propose two hypotheses:

H_0: Model (1) is the correct model.

H_1: Model (2) is the correct model.

Under the null hypothesis, all model coefficients are fixed, including $S_Risk\text{-}taking_{it}$, which does not change with Z_{it}. In other words, the moderating effect variable Z_{it} is linear. The alternative hypothesis (H_1) is that Model (2) is the correct model, suggesting that the relationship between $S_Risk\text{-}taking_{it}$ and C_ESG_{it} is nonlinear and varies with Z_{it}.

To test these hypotheses, we develop the T-statistic, defined as follows:

$$T = \frac{\sum \widehat{\varepsilon}_{1,it} - \sum \widehat{\varepsilon}_{2,it}}{\sum \widehat{\varepsilon}_{2,it}} \tag{3a}$$

where $\sum \widehat{\varepsilon}_{1,it}$ is the residual sum of squares of Model (1), $\sum \widehat{\varepsilon}_{2,it}$ is the residual sum of squares of Model (2).

Following prior studies (Kreiss et al. (2008), Cai (2007) and Cai et al. (2000)), we use the non-parametric wild bootstrap method to directly obtain the sample distribution of T. We compute 1000 bootstrap statistics to determine the 1 %, 5 %, and 10 % critical values of the empirical distribution of the bootstrapped statistic. If the p-value is less than 10 %, we reject the null hypothesis and conclude that Model 2 (the partially linear functional-coefficient panel data model) is the correct model.

The testing method employed in this study has the following three key features: First, the statistic constructed uses the partially linear functional-coefficient model and the linear interaction term panel data model effectively tests whether the impact of the core explanatory variable on the dependent variable is influenced by the non-linear effect of the moderating variable. Second, during the wild bootstrap process, the residuals from the model estimation are used. Regardless of whether the null hypothesis holds, the residuals we use are asymptotically consistent, which improves the power of our test statistic. Third, by calculating the p-value through wild bootstrap, we ensure that the test maintains favorable large-sample properties when using non-parametric estimation in panel data. As shown in Appendix Table 1, all the T-statistics are statistically significant at the 1 % (or 5 %) level across all estimations, suggesting that the partially linear functional-coefficient panel data model is statistically superior to the linear interaction term panel data model at the 1 % (or 5 %) significance level.

Appendix Table 1
Functional form test of partially linear functional-coefficient model

Dependent variable: C_ESG			
	(1)	(2)	(3)
Moderator	*Supplier dependence*	*Corporate reputation*	*Investor confidence*
T statistic	−0.127***	−0.196***	−0.130***
	(0.000)	(0.000)	(0.000)
	(4)	(5)	(6)
Moderator	*Corporate Digitalization*	*Artificial intelligence*	*Other digital technology*
T statistic	−0.131***	−0.290**	−0.129***
	(0.000)	(0.035)	(0.000)

Notes: * < 0.1, ** < 0.05, *** < 0.01.P-values are shown in brackets.

Data availability

Data will be made available on request.

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